

RetinAnalysis Setup — Windows

Everything runs in **Anaconda PowerShell Prompt**. No Git Bash needed anywhere in this guide.

Skip a step if you already have that tool installed. Each step says what it's for so you can tell.

1. Install Git

Needed to clone the repository. Download and run the installer, accepting the defaults:

<https://git-scm.com/download/win>

This installer also bundles Git Bash — you don't need to use it. After installing, `git` itself works directly from Anaconda PowerShell Prompt.

2. Install Docker Desktop

Runs the local MySQL/DataJoint database that stores experiment metadata. Download and install, then launch it once:

<https://docs.docker.com/desktop/>

Leave Docker Desktop open whenever you need database access (queries, `ra.populate_database()`). Just `import retinanalysis` does not need it running.

3. Install a C++ compiler

One required piece of this package (the `vision-utils` submodule, which reads Vision's binary recording files) compiles C++ extensions the first time it's installed — this needs an actual compiler present, or that install step will fail.

<https://visualstudio.microsoft.com/visual-cpp-build-tools/>

In the installer, check **"Desktop development with C++"**, then install (a few GB, may prompt a restart). You don't need full Visual Studio or an IDE — the Build Tools package alone is enough.

4. Confirm you have conda

Open **Anaconda PowerShell Prompt** and run:

```
conda --version
```

If that fails, install Miniconda first: <https://docs.conda.io/en/latest/miniconda.html>

5. Clone the repository

`--recursive` is required — it pulls in a submodule (`vision-utils`) that the package needs to read Vision's data files.

```
git clone https://github.com/yasmine-tani/retinanalysis.git --recursive
cd retinanalysis
```

If you forget `--recursive`, fix it afterward with `git submodule update --init --recursive`.

6. Create the environment and install

```
conda create -y -n retinanalysis python=3.11.13
conda activate retinanalysis
pip install -e .
pip install .\lib\artificial-retina-software-pipeline\utilities\
```

The first `pip install` installs retinanalysis itself and everything it depends on (including `opencv-python` — you never need to install OpenCV separately). The second installs the submodule from step 5 — this is the step that needs the C++ compiler from step 3.

7. Fix a known Windows bug

The first time this package tries to load `matplotlib`, Windows can raise a DLL load error. If you hit that, fix it with:

```
pip uninstall Pillow
pip install -U Pillow
```

If you don't hit the error, skip this — it's not something to run preemptively.

8. Set up config.ini

This file tells the package where your data actually lives on this machine. It doesn't exist yet — every machine's paths are different, so you're creating it fresh. Create a new file at

`src\retinanalysis\config\config.ini`, paste in the block below, then edit the

`[WINDOWS_DEFAULT]` section with real paths:

```
[WINDOWS_DEFAULT]
analysis = B:\Array-data\sorted
data = B:\Array-data\sorted
raw = B:\Array-data\raw
h5 = B:\Array-data\h5
meta = B:\Array-data\dj_meta
tags = B:\Array-data\tags
query = B:\Array-data\sorted
user = your_username

[WINDOWS_SECONDARY]
analysis = C:\path\to\secondary\analysis
data = C:\path\to\secondary\sorted
raw = C:\path\to\secondary\raw
h5 = C:\path\to\secondary\h5
meta = C:\path\to\secondary\meta
tags = C:\path\to\secondary\tags
query = C:\path\to\secondary\query
user = your_username
```

`B:` is whatever drive letter your shared network drive is mapped to on this machine — check File Explorer → This PC. `user` is your own username. Both sections must be present in the file (the package reads both on startup), but you only need to fill in real paths under `[WINDOWS_DEFAULT]` — leave `[WINDOWS_SECONDARY]` as placeholder text unless you actually have a second data location; a path that doesn't exist is just skipped.

9. Start the local database

Docker Desktop must be open for this to work.

```
mkdir ..\retinanalysis-database
cp docker-compose.yaml ..\retinanalysis-database\
cd ..\retinanalysis-database
docker compose up -d
cd ..\retinanalysis
```

10. Install Jupyter and launch it

```
pip install jupyterlab
jupyter lab
```

This opens a browser tab. Because Jupyter was installed inside the same `retinanalysis` environment you just activated, it's automatically using the right Python — no separate kernel setup needed.

11. Populate the database

Open any notebook in `demos/` and run in a cell:

```
import retinanalysis as ra
ra.populate_database()
```

Quick reference — every command in order

```
git clone https://github.com/yasmine-tani/retinanalysis.git --recursive
cd retinanalysis
conda create -y -n retinanalysis python=3.11.13
conda activate retinanalysis
pip install -e .
pip install .\lib\artificial-retina-software-pipeline\utilities\
REM create src\retinanalysis\config\config.ini (see step 8), then:
mkdir ..\retinanalysis-database
cp docker-compose.yaml ..\retinanalysis-database\
cd ..\retinanalysis-database
docker compose up -d
cd ..\retinanalysis
pip install jupyterlab
jupyter lab
```

Troubleshooting

ModuleNotFoundError: No module named 'cv2' on the very first `import retinanalysis` cell

Not a missing OpenCV/C++ install — `opencv-python` is already installed automatically by step 6. This means Jupyter is running a different Python than the `retinanalysis` environment. Close Jupyter, run `conda activate retinanalysis`, then launch `jupyter lab` from that same terminal. Once the notebook is open, check the kernel name in the top-right corner — if it's not `retinanalysis`, switch it (or if it's missing from the list, run `pip install ipykernel` then `python -m ipykernel install --user --name retinanalysis` and refresh).

DLL load failed on `import matplotlib`

See step 7.

FileNotFoundError: No config file found at ... on `import retinanalysis`

Step 8 was skipped, or the file isn't named/placed exactly right. It must be at `src\retinanalysis\config\config.ini` (not `.txt`, not `config.ini.txt` — make sure File Explorer isn't hiding the real extension). Create it as described in step 8, then restart the kernel and re-run.

No NAS or SSD paths found, check that one of them is connected printed but no crash

`config.ini` exists but still has placeholder paths (e.g. `B:\Array-data\sorted`) that don't actually exist on this machine. Open the file and confirm the `[WINDOWS_DEFAULT]` paths point to a drive/folder that's really there — check File Explorer → This PC for the right drive letter.

Installing the `lib\...\utilities\` submodule fails with a compiler error

The C++ Build Tools from step 3 aren't installed, or you need a brand-new PowerShell window opened after installing them.

Database calls fail / connection refused

Docker Desktop isn't open, or the container isn't running. Open Docker Desktop and confirm the `retinanalysis-database` container shows a stop icon (running), not a play icon.

Cloned without `--recursive`

Run `git submodule update --init --recursive` , then redo step 6.